

PAPER_1404_SOLAR_NEUTRINO_DEFICIT

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PAPER_1404 — UQFF Derivation of Solar Neutrino Deficit (CLOSED — EXACT Identity)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: D — Particle Physics (CLOSED — EXACT)

Date: June 16, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — EXACT integer-primitive identity

Calculator surface: `calculate_particle_physics({'observable':'solar_neutrino'}) → 1/(D_phys-1) = 1/3 EXACT`

Observation

The historic 1968 Homestake solar ν_μ deficit (only $\sim 1/3$ of predicted flux observed) is resolved by 3-flavor neutrino oscillation — but the **1/3** ratio is itself unexplained. UQFF derives it as a pure integer-primitive identity.

UQFF Closed Identity

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$P(\nu_\mu \mid \text{observed}) = 1 / (D_{\text{phys}} - 1) = 1/3 \text{ EXACT}$

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Physical Interpretation

$D_{\text{phys}} = 4$ spacetime dimensions implies 3 lepton generations; $F_U=1$ ledger normalization redistributes the original ν_μ flux equally across all 3 flavors, giving $1/3$ detected. The deficit was a clue to $D_{\text{phys}}-1 = 3$.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve the same observation by different methods. UQFF derives the result above using **only canonical integer primitives** $\{D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60\}$ and locked real primitives. **Zero free parameters. EXACT identity.**

Reference

- Closure live in `uqff_pure_calculator.py` at the catalog surface listed above.
- Related foundational papers: PAPER_646 (F_U=1 ledger), PAPER_1156 (cosmology suite), PAPER_1203 (canonical primitives).

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, dated June 16, 2026, location 41.0997° N, 80.6495° W (Youngstown, OH, USA). Subject matter: UQFF EXACT integer-primitive identity for Solar Neutrino Deficit.